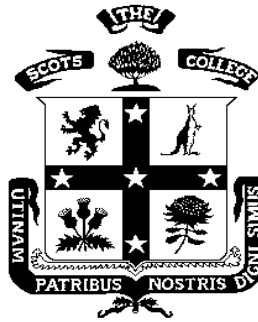


The Scots College



Year 12 Mathematics Extension 2

Pre-Trial Assessment

April 2009

General Instructions

- All questions are of equal value
- Working time - 2 hours
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Standard Integrals Table is attached

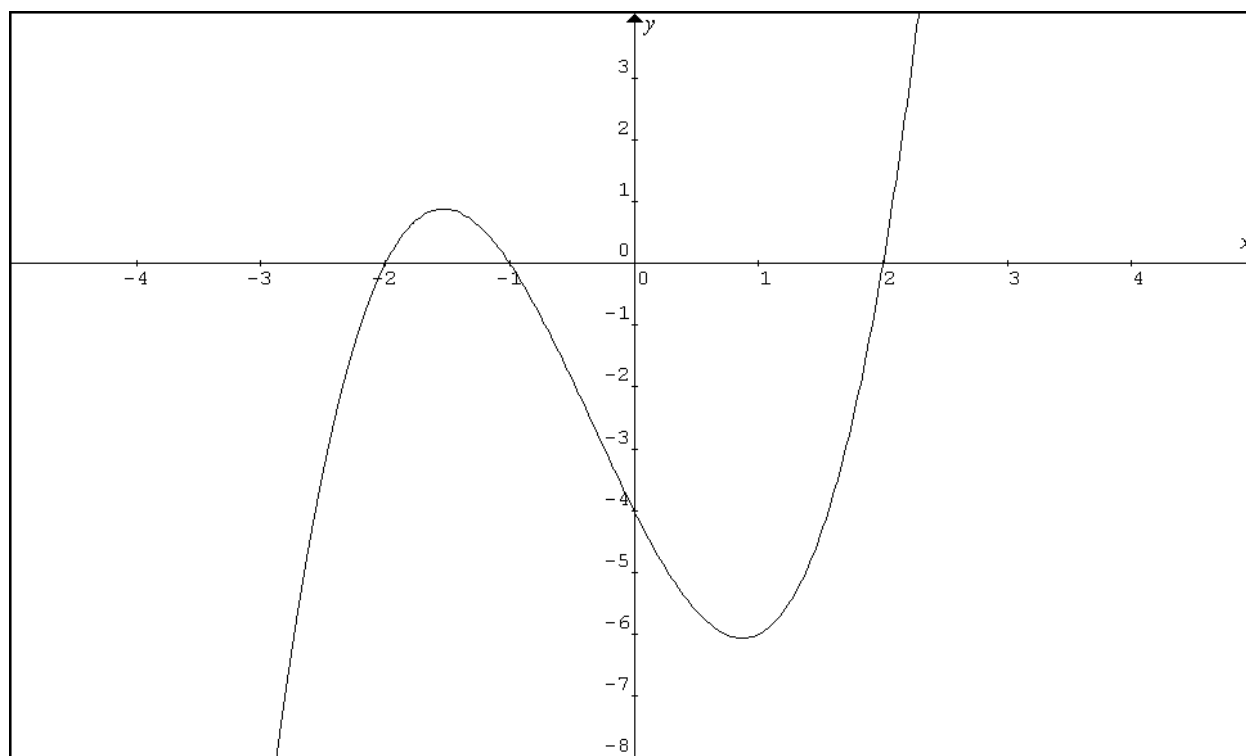
TOTAL MARKS: 75

WEIGHTING: 30 %

- **Start each question in a new booklet**

QUESTION 1**[15 MARKS]**

- (a) In the diagram below, the graph $f(x)$ is drawn.



On separate diagrams, and without the use of calculus, sketch the following graphs. Indicate clearly any asymptotes, any intercepts with the coordinate axes, and any other significant features.

(i) $y = \frac{1}{f(x)}$ 2

(ii) $y = -\sqrt{f(x)}$ 2

(iii) $y = e^{f(x)}$ 2

(iv) $y = f(2-x)$ 2

(v) $y = f'(x)$ 2

- (b) Given the function $f(x) = \sqrt{2 - \sqrt{x}}$.

(i) What is the domain of this function 1

(ii) Show that $f(x)$ is a decreasing function and hence find its range 1

(iii) Draw a neat sketch of $f(x) = \sqrt{2 - \sqrt{x}}$. 3

QUESTION 2**[15 MARKS]**

(a) Find $\int 7x\sqrt{(4x^2 - 3)} dx$ 2

(b) Find $\int \frac{x+1}{\sqrt{(x-1)}} dx$ 2

(c) Evaluate

(i) $\int_0^{\frac{\pi}{3}} \tan x \cdot \sec^4 x dx$ 2

(ii) $\int_0^{\frac{\pi}{2}} \frac{dx}{\cos x + 2}$, by using the substitution $t = \tan \frac{x}{2}$ 2

(c) Find the values of A and B such that

$$\frac{e^x}{(e^x + 2)(e^x + 1)} = \frac{A}{(e^x + 2)} + \frac{B}{(e^x + 1)}$$
 1

Hence find $\int \frac{e^x}{e^{2x} + 3e^x + 2} dx$. 2

(d) $I_n = \int_0^{\frac{\pi}{2}} \cos^n x dx \quad n \geq 0$

(i) Show that $I_n = \frac{n-1}{n} I_{n-2}$ 2

(ii) Hence evaluate $\int_0^{\frac{\pi}{2}} \cos^4 x dx$ 2

QUESTION 3**[15 MARKS]**

- (a) Find the equation of the circle which has the points A (3 , -1) and B (9 , 3) which are at opposite ends of a diameter. **2**
- (b) State the foci, directrices and eccentricity of $4x^2 + 25y^2 - 100 = 0$ **2**
- (c) Prove that for any point P on the Hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, the **difference** of its distances from the foci, S and S' is constant. **2**
- Find the value of this constant. **1**
- (d)

QUESTION 4**[15 MARKS]**

- (a) Reduce the complex number $(2 - i)(8 + 3i)/(3 + i)$ to the form $a + ib$, where a and b are real numbers. 2
- (b) The complex number z is given by $z = -\sqrt{3} + i$.
- (i) Write down the values of $\text{Arg}(z)$ and $|z|$ 1
- (ii) Hence or otherwise show that $z^7 + 64z = 0$ 2
- (c) Sketch the following loci on separate Argand diagrams;
- (i) $\text{Arg}(z + 1 + i) = \pi/4$ 1
- (ii) $|z - 2| = |z + i|$ 1
- (d) Given that $z = 3 + 4i$. Find w so that O , z and w form a right angled isosceles triangle, (whose right angle is at z) on the Argand diagram. 3
- (e) Using any complex number z and on separate Argand diagrams, illustrate the geometric properties of the following;
- (i) z^2 1
- (ii) $z \times (1 + i)$ 1
- (iii) $z - \bar{z}$ 1
- (f) Given $z^4 = i$. Find the four roots of unity 2

QUESTION 5**[15 MARKS]**

(a) Using Integration by parts, to show $\int_1^e \sin(\ln x) dx = \frac{e}{2}(\sin 1 - \cos 1) + \frac{1}{2}.$

4

(b) **Complex qu**

(c) Sketch the following equation; $|y| = (x-1)(x-2)(x-3)$

4

(d) **Conics qu**

END OF PAPER

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

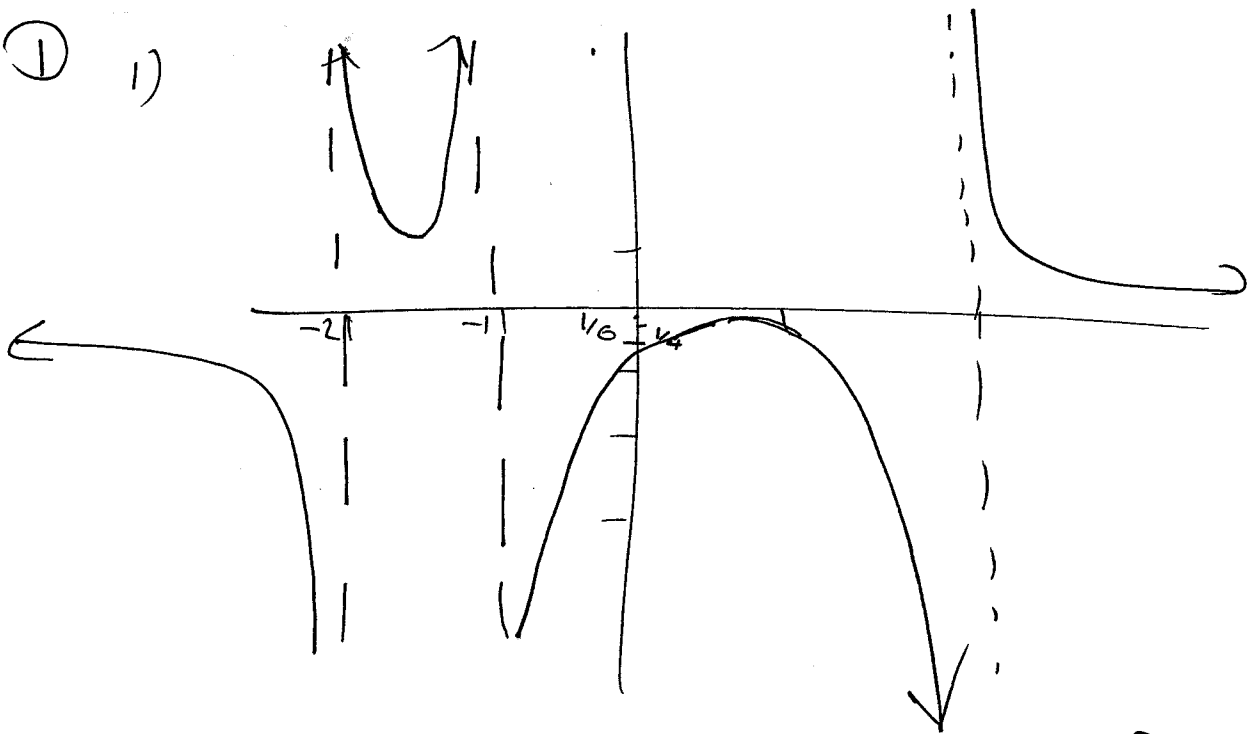
$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

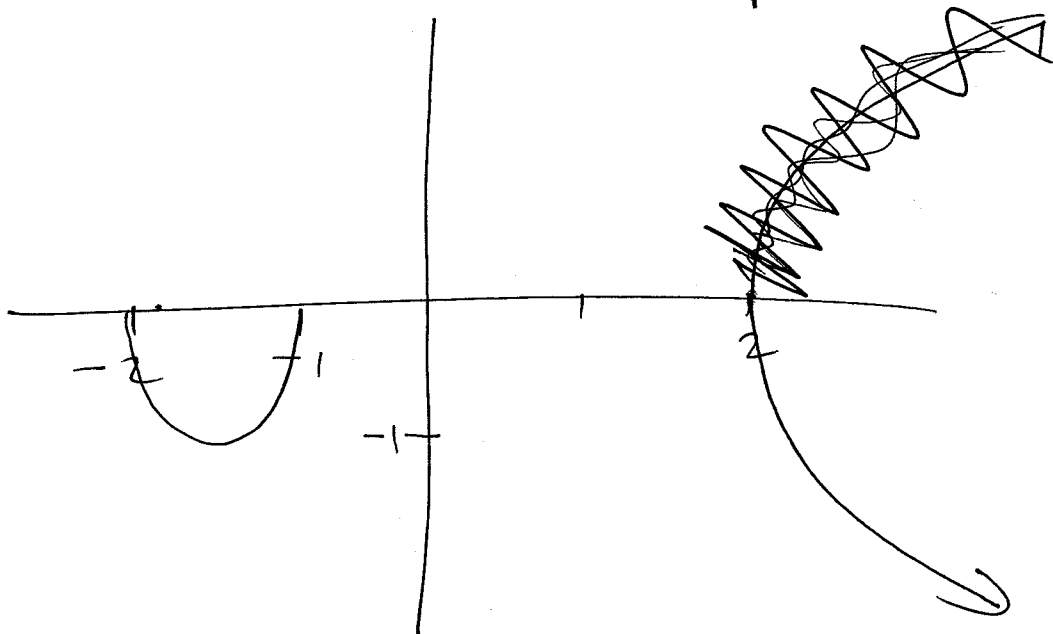
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE : $\ln x \equiv \log_e x, \quad x > 0$

① 1)

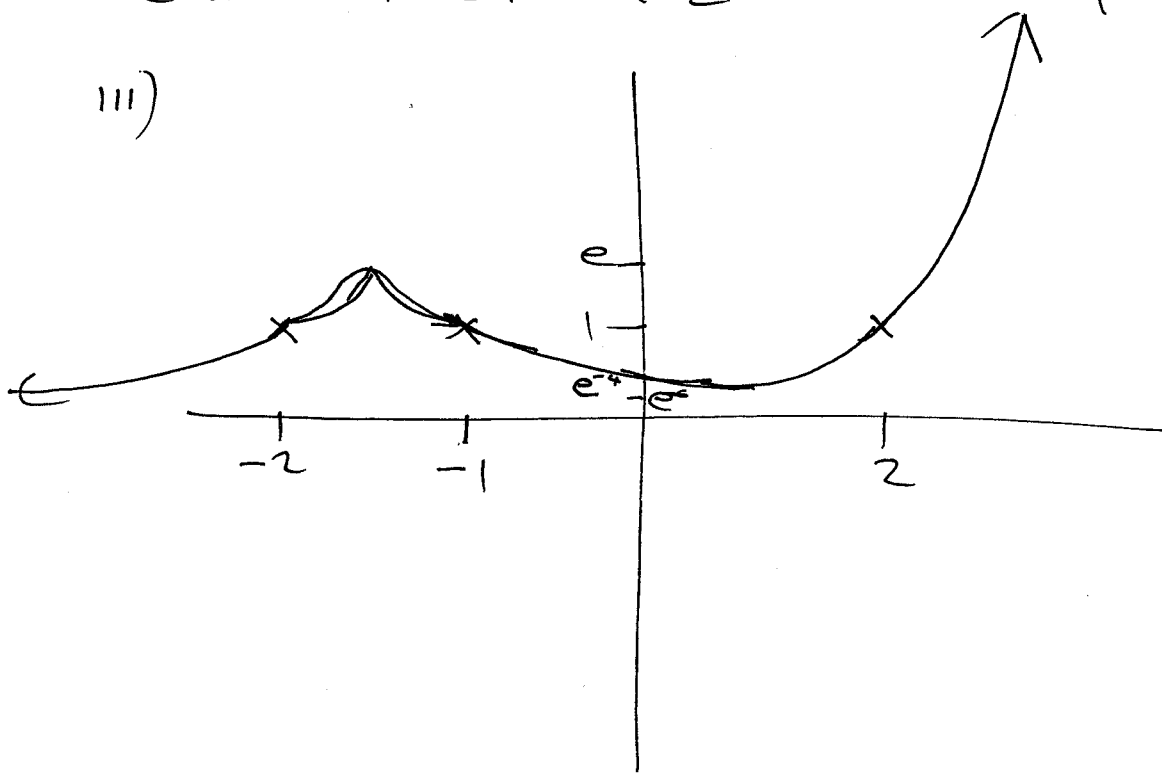


11)

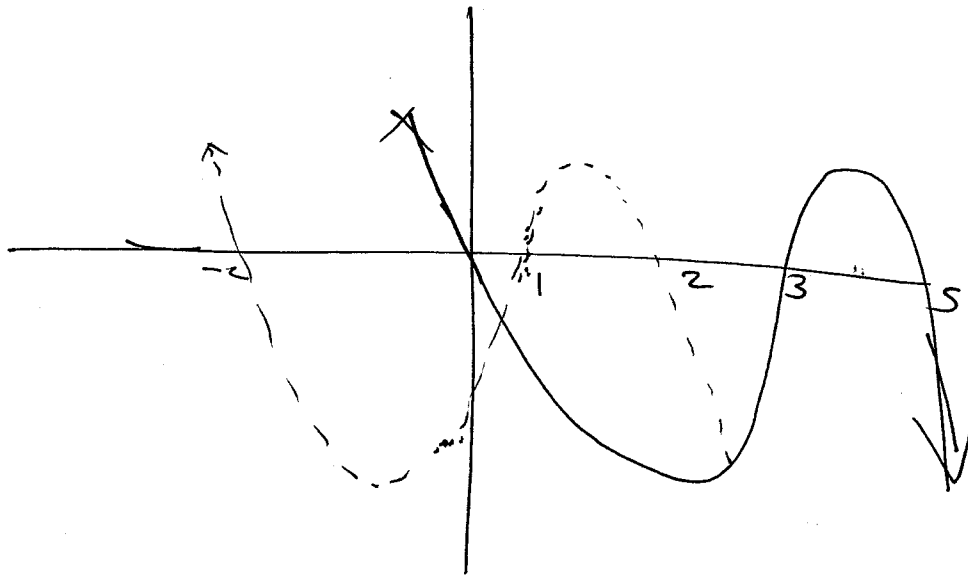


111)

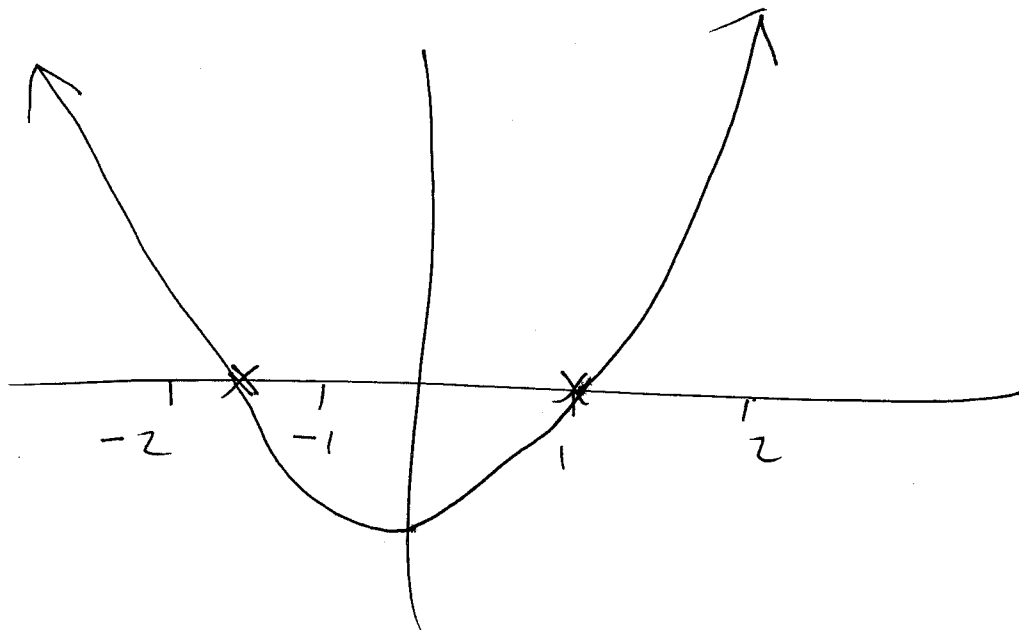
III)



IV)



V)



$$b) f(x) = \sqrt{2-\sqrt{x}}$$

i) Domain $\sqrt{x} \leq 2$

$$0 \leq x \leq 4$$

$$ii) f(x) = (2 - x^{1/2})^{1/2}$$

$$\therefore f'(x) = +\frac{1}{2}(2 - x^{1/2})^{-1/2} \cdot -\frac{1}{2}x^{-1/2}$$

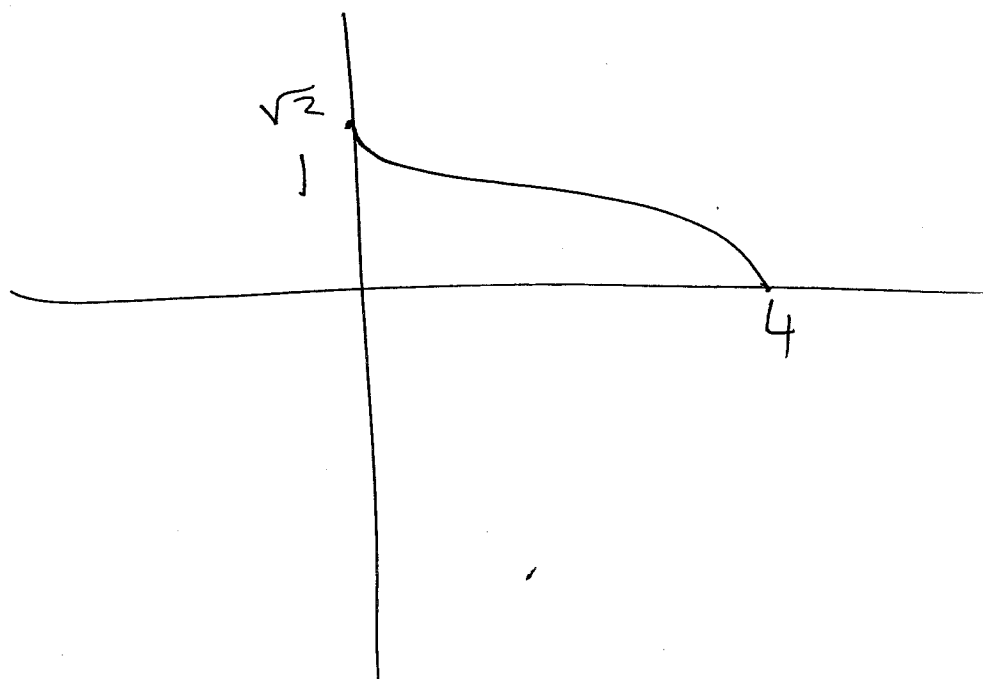
$$= -\frac{1}{4(2 - x^{1/2})^{1/2} x^{1/2}} \leftarrow \text{Positive for Domain}$$

$\therefore f'(x)$ always -ve $\therefore$ Always decrease

$$\therefore f(0) = \sqrt{2} \quad f(4) = 0$$

$$\therefore \text{Range } 0 \leq y \leq \sqrt{2}$$

iii)



a)

$$\int 7x\sqrt{4x^2-3}$$

by observation $7x(4x^2-3)^{1/2}$

$$\frac{7}{12}(4x^2-3)^{3/2} + C$$

b)

$$\int \frac{x+1}{\sqrt{x-1}} dx \quad \text{by sub } u=x-1$$

$$\frac{du}{dx} = 1$$

$$du = dx$$

$$\int \frac{u+2}{\sqrt{u}} du$$

$$\int u^{1/2} du + \int 2u^{-1/2} du$$

$$= \frac{2}{3}u^{3/2} + 4u^{1/2} + C$$

Can tidy up

$$\frac{2}{3}(x-1)^{3/2} + 4(x-1)^{1/2}$$

$$(x-1)^{1/2} \left[\frac{2}{3}(x-1) + 4 \right]$$

$$\frac{1}{3}(x-1)^{1/2} (2x-10) + C$$

C

$$\int_0^{\pi/3} \tan x \sec^4 x dx$$

$$\tan x \sec^2 x \sec^2 x$$

$$\tan x (\tan^2 x + 1) \sec^2 x$$

$$\sec^2 x \tan^3 x + \sec^2 x \tan x$$

$$\left[\frac{1}{4} \tan^4 x + \frac{1}{2} \tan^2 x \right]_0^{\pi/3}$$

$$\frac{1}{4} (\sqrt{3})^4 + \frac{1}{2} (\sqrt{3})^2$$

$$\frac{9}{4} + \frac{3}{2} = 3\frac{3}{4}$$

$$\int_0^{\pi/2} \frac{dx}{\cos x + 2}$$

$$\int_0^1 \frac{1}{\frac{1-t^2}{1+t^2} + 2} \times \frac{2}{1+t^2} dt$$

$$\int_0^1 \frac{1}{\frac{1-t^2+2+2t^2}{1+t^2}} dt$$

$$\int_0^1 \frac{\cancel{1+t^2} \times 2}{3+t^2} dt$$

$$\frac{2}{\sqrt{3}} \int_0^1 \frac{1}{3+t^2} dt \Rightarrow \frac{2}{\sqrt{3}} \left[\tan^{-1} \frac{t}{\sqrt{3}} \right]_0^1$$

$$\frac{2}{\sqrt{3}} \left[\tan^{-1} \frac{1}{\sqrt{3}} \right] - 0$$

$$\frac{2}{\sqrt{3}} \frac{\pi}{3\sqrt{3}}$$

$$t = \tan \frac{x}{2}$$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$\frac{dx}{dt} = \frac{2}{\sec^2 \frac{x}{2}}$$

$$dx = \frac{2}{1+t^2} dt$$

$$x = \frac{\pi}{2} \quad t = \tan \frac{\pi}{4}$$

$$x = 0 \quad t = \tan 0$$

d) $e^x = A(e^x+1) + B(e^x+2)$

$$e^x = Ae^x + Be^x + A + 2B$$

$$\therefore A+B=1$$

$$A+2B=0$$

$$A=1-B$$

$$1-B+2B=0$$

$$\begin{aligned} B &= -1 \\ A &= 2 \end{aligned}$$

$$\int \frac{2}{e^x+2} dx = \int \frac{1}{e^x+1} dx$$

~~or~~

let $e^x+2=U$

$$\therefore \frac{du}{dx} = e^x$$

$$dx = \frac{du}{e^x}$$

$$\int \frac{2}{U} \times \frac{du}{U-2}$$

$$\int \frac{2}{U^2} = \frac{2}{U(U-2)} = \frac{A}{U} + \frac{B}{(U-2)}$$

$$2 = A(U-2) + BU$$

$$U=0 \quad A=-1$$

$$U=2 \quad B=1$$

$$\int -\frac{1}{U} + \frac{1}{U-2}$$

let $e^x+1=V$

$$\frac{dV}{dx} = e^x$$

$$dx = \frac{dV}{e^x}$$

$$\int \frac{1}{V} \times \frac{dV}{V-1}$$

$$\frac{1}{V(V-1)} = \frac{A}{V} + \frac{B}{V-1}$$

$$1 = A(V-1) + BV$$

$$V=0 \quad A=-1$$

$$V=1 \quad B=1$$

$$\int -\frac{1}{V} + \frac{1}{V-1}$$

$$\begin{aligned}
 & -\ln U + \ln U - 2 \\
 & \ln \frac{U-2}{U} - \ln U \\
 & \ln e^x - \ln e^{x+2} \\
 & 1 - \ln(e^x + 2)
 \end{aligned}$$

$$\begin{aligned}
 & -\ln V + \ln V - 1 \\
 & \ln V - 1 - \ln V \\
 & \ln e^x - \ln e^{x+1} \\
 & 1 - \ln(e^x + 1)
 \end{aligned}$$

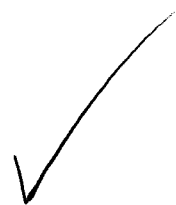
$$\ln(e^x + 1) - \ln(e^x + 2)$$

$$e^x$$

$$\frac{(e^x + \frac{3}{2})^2 - \frac{1}{4}}{e^x}$$

$$\frac{e^x}{e^{x+1}} - \frac{e^x}{e^{x+2}}$$

$$\begin{aligned}
 & \frac{\cancel{e^{2x}} + 2e^x - \cancel{e^{2x}} - e^x}{() ()} \\
 & \frac{e^x}{() ()}
 \end{aligned}$$



Ext 2 tutorial 2009 - question 2

$$I_n = \int_0^{\frac{\pi}{2}} \cos^n x dx \quad n \geq 0$$

$$\int_0^{\frac{\pi}{2}} \cos^2 x + \cos^{n-1} x dx$$

$$dv = \cos x \quad u = \cos^{n-1} x$$

$$v = \sin x \quad du = (n-1) \cos^{n-2} x (-\sin x)$$

$$I_n = \left[\sin x \cos^{n-1} x \right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} (n-1) \sin^2 x \cos^{n-2} x dx$$

$$= (n-1) \int_0^{\frac{\pi}{2}} (1 - \cos^2 x) \cos^{n-2} x dx$$

$$= (n-1) \int_0^{\frac{\pi}{2}} \cos^{n-2} x dx - (n-1) \int_0^{\frac{\pi}{2}} \cos^n x dx$$

$$= (n-1) I_{n-2} - (n-1) I_n$$

$$n I_n = (n-1) I_{n-2}$$

$$I_n = \frac{(n-1)}{n} I_{n-2}$$

$$I_4 = \frac{3}{4} I_2$$

$$I_2 = \frac{1}{2} I_0$$

$$I_0 = \int_0^{\frac{\pi}{2}} \cos^0 x dx$$

$$= \left[x \right]_0^{\frac{\pi}{2}}$$

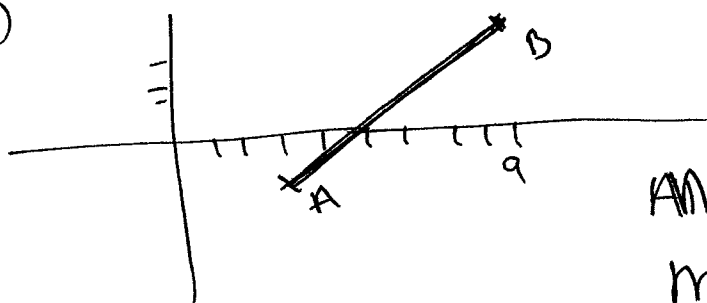
$$= \frac{\pi}{2}$$

$$I_2 = \frac{\pi}{4}$$

$$I_4 = \frac{3\pi}{16}$$

//

a)



Mid Point = Centre

(6, 1)

$$AB^2 = 3^2 + 2^2$$

$$MB^2 = 13 = \text{Radii}^2$$

$$\therefore (x-6)^2 + (y-1)^2 = 13$$

b)

$$\frac{x^2}{5^2} + \frac{y^2}{2^2} = 1 \quad [\text{Ellipse}]$$

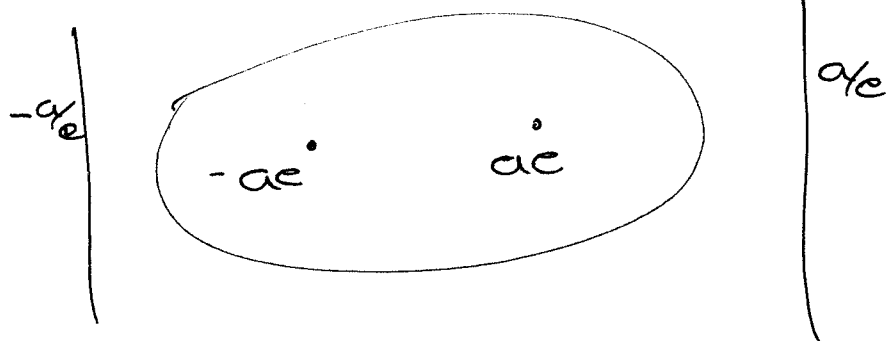
$$b^2 = a^2(1-e^2)$$

$$\frac{4}{25} = 1 - e^2$$

$$\therefore e^2 = \frac{21}{25}$$

$$e = \frac{\sqrt{21}}{5}$$

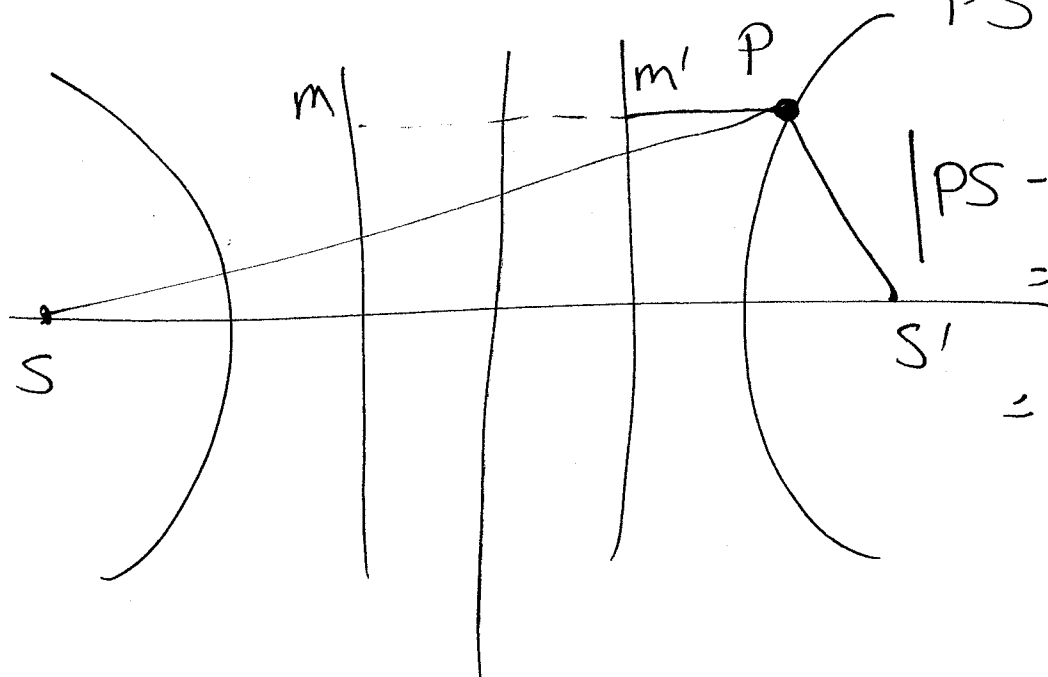
c)



$$\text{Foci } [\pm \sqrt{21}, 0]$$

$$\text{Direct: } [x = \pm \frac{25}{\sqrt{21}}]$$

c)



$$PS' = ePM'$$

$$PS = ePM$$

$$= e[PM' + 2ae]$$

$$|PS - PS'|$$

$$= e[PM' - PM' + 2ae]$$

$$= 2a$$

$y = \frac{c^2}{x}$
 $\frac{dy}{dx} = -\frac{c^2}{x^2}$
 $\text{at } p = -\frac{c^2}{c^2 p^2}$
 $M_{\text{tan}} = -\frac{1}{p^2}$
 $M_{\text{normal}} = p^2$

$y - \frac{c}{p} = p^2(x - cp)$
 $yp - c = p^3(x - cp)$

ii) For this to be true

$$xy = c^2 \quad \text{--- (1)}$$

$$py - c = p^3(x - cp)$$

Rearrange (1)

$$y = \frac{c^2}{x}$$

$$py - c = \frac{pc^2}{x} - c$$

$$\therefore \frac{pc^2}{x} - c = p^3(x - cp)$$

$$\frac{c^2}{p^2} - \frac{cx}{p^3} = x^2 - cp^2x$$

$$x^2 + \frac{c}{p^3}x - cp^2x - \frac{c^2}{p^2} = 0$$

$$x^2 - c\left[p - \frac{1}{p^3}\right]x - \frac{c^2}{p^2} = 0$$

$\therefore$ P and Q satisfy the equation

Roots of Quadratic are X_Q, X_P

$$\therefore \text{Product of Roots} = \frac{C^2}{p^2}$$

$$X_P = Cp$$

$$X_Q = \frac{C}{p^3}$$

Sub in $XY = C^2$

$$Y_Q = \frac{C^2}{C/p^3}$$

$$Y_Q = Cp^3$$

$$Q \left(-\frac{C}{p^3}, Cp^3 \right)$$

$XY = C^2$ is odd function

$$\therefore R \left(-Cp, \frac{C}{p} \right)$$

$$M_{PR} = \left[\frac{\frac{2C}{p}}{\frac{2Cp}{p^2}} \right]$$

$$= \frac{1}{p^2}$$

$$M_{PR} \left[\frac{-Cp^3 - \frac{C}{p}}{-\frac{C}{p^3} - Cp} \right]$$

$$= \frac{C}{C} \left[\frac{\frac{p^4 - 1}{p}}{\frac{p^4 - p^4}{p^3}} \right]$$

$$= -p^2$$

$$\frac{1}{p^2} - p^2 = -1$$

$$\therefore \underline{\underline{H}}$$

a) $\frac{5+i}{3-2i} \times \frac{3+2i}{3+2i}$

$$\frac{15 + 10i + 3i - 2}{9 + 4}$$

$$\frac{13 + 13i}{13}$$

$$1 + i$$

$\sqrt{2} \text{ CLS } \pi/4$

b) $w^2 = 5 - 12i$

$$(a+bi)^2 = 5 - 12i$$

$$a^2 - b^2 + 2abi = 5 - 12i$$

$$a^2 - b^2 = 5 \quad 2ab = -12$$

$$a = \frac{-6}{b}$$

$$\frac{36}{b^2} - b^2 = 5$$

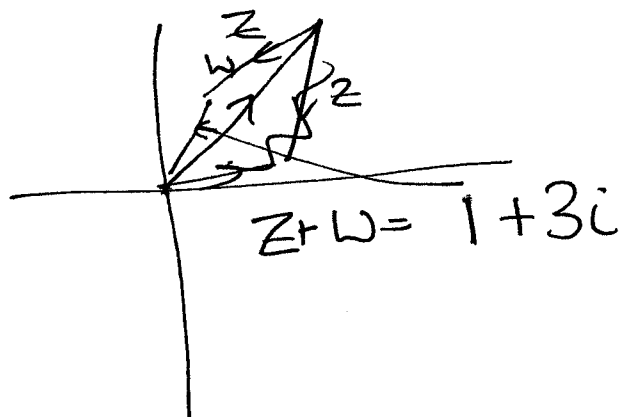
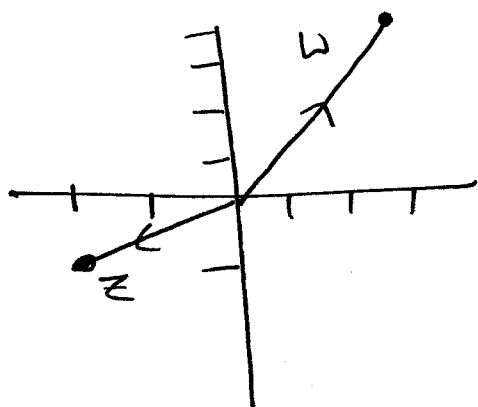
$$b^4 + 5b^2 - 36 = 0$$

$$[b^2 + 9][b^2 - 4] = 0$$

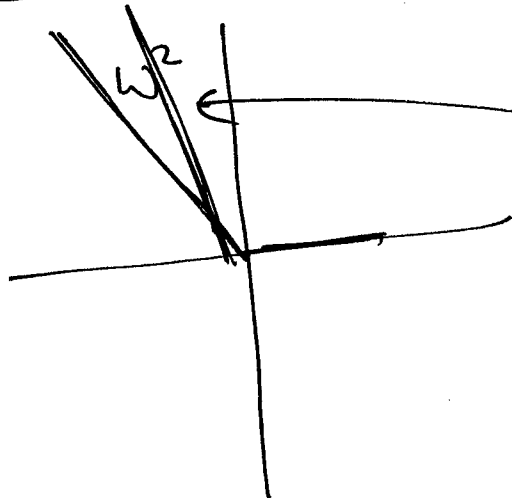
Reject $b = \pm 2$

$a = \mp 3$

$(-3+2i) \quad (3-2i)$



$$z \cdot \bar{z} + w^2$$



$$w^2 = 9 - 16 + 24i$$

$$\therefore -7 + 24i$$

$$z\bar{z} = (-2-i)(-2+i)$$

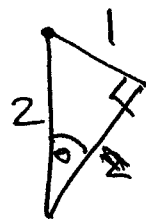
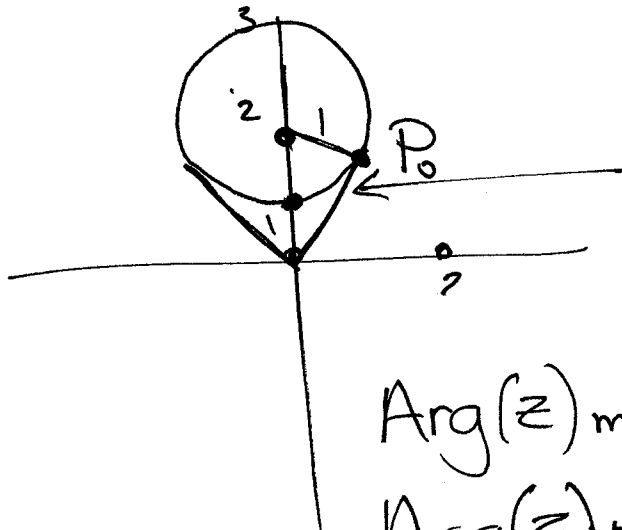
$$4 + 1 = 5$$

$-2 + 24i$

Ext. the max. min.

position ~~of~~

d)



$$\sin \theta = \frac{1}{3}$$

$$\frac{\pi}{6}$$

$$\text{Arg}(z)_{\min} \frac{\pi}{3}$$

$$\text{Arg}(z)_{\max} \frac{2\pi}{3}$$

$$P_0 = \sqrt{3} \text{ CIS } \frac{\pi}{3}$$

e)

$$z^6 = -1$$

$$z^6 = 1 \text{ CIS } -\frac{\pi}{2}$$

$$\therefore z_1 = 1 \text{ CIS } -\frac{\pi}{12}$$

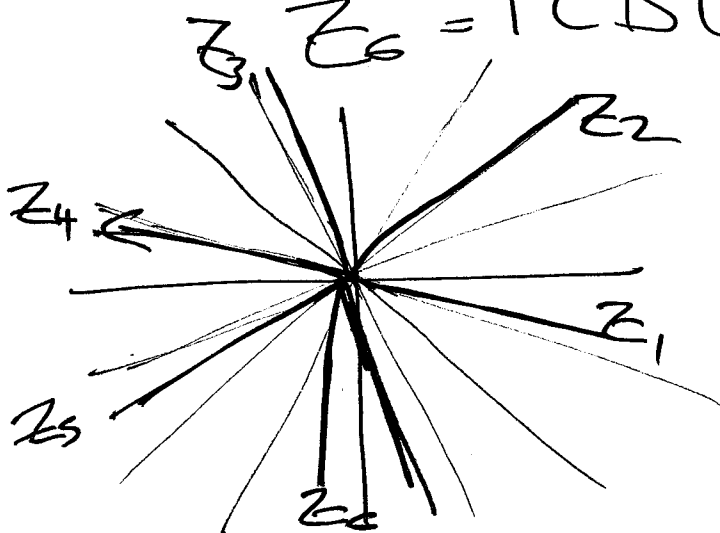
$$z_2 = 1 \text{ CIS } \left(-\frac{\pi}{12} + \frac{\pi}{3}\right) = \text{CIS} \left(\frac{3\pi}{12}\right) = \text{CIS} \frac{\pi}{4}$$

$$z_3 = 1 \text{ CIS } \left(\frac{7\pi}{12}\right)$$

$$z_4 = 1 \text{ CIS } \left(\frac{11\pi}{12}\right)$$

$$z_5 = 1 \text{ CIS } \left(\frac{15\pi}{12}\right) = 1 \text{ CIS } \left(-\frac{9\pi}{12}\right) = \text{CIS} \left(-\frac{3\pi}{4}\right)$$

$$z_6 = 1 \text{ CIS } \left(-\frac{5\pi}{12}\right)$$



Multiply to make
an imaginary number
no conjugate

$$a) \int_1^e \sin(\ln x) dx$$

$$dv=1 \quad u=\sin(\ln x)$$

$$v=x \quad du = \frac{1}{x} \cos(\ln x)$$

$$I = [x \sin(\ln x)]_1^e + \int_1^e -\cos(\ln x) dx$$

$$\frac{dv}{dx}=1 \quad u=-\cos(\ln x)$$

$$v=x \quad du = \frac{1}{x} \sin(\ln x)$$

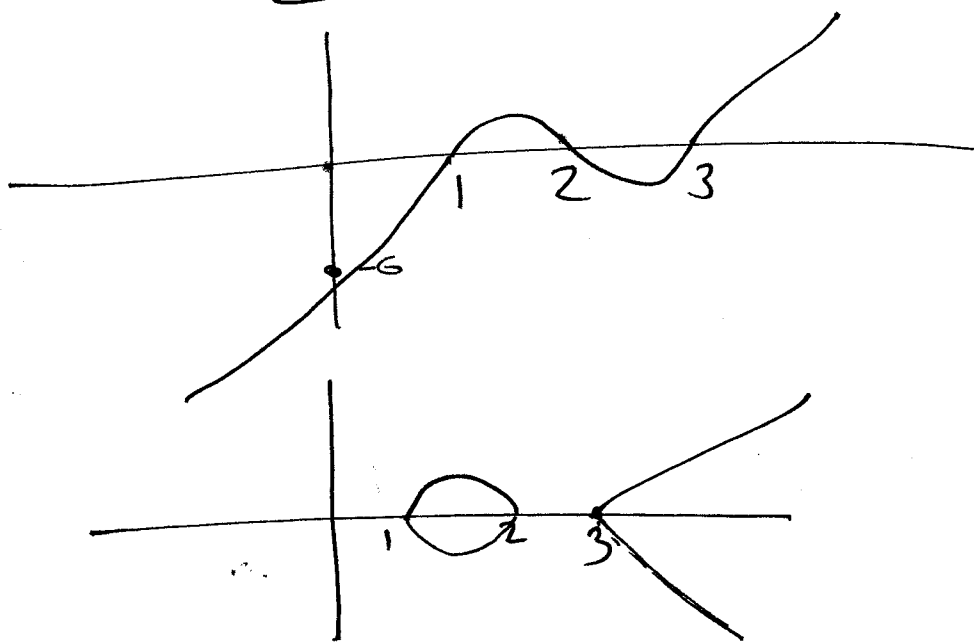
$$I = [x \sin(\ln x)]_1^e + [-x \cos(\ln x)]_1^e - I$$

$$2I = e \sin 1 - 0 + -e \cos 1 + 1$$

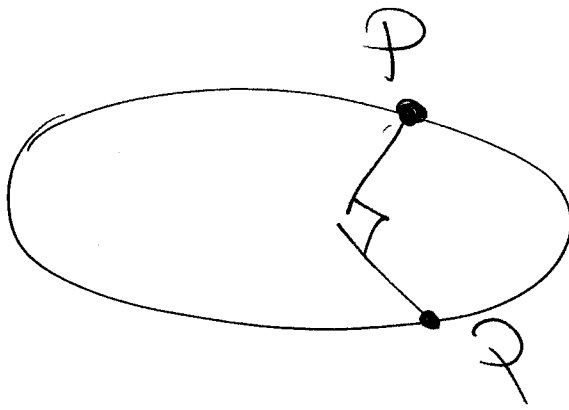
$$2I = e [\sin 1 - \cos 1] + 1$$

$$I = \frac{e}{2} [\sin 1 - \cos 1] + \frac{1}{2}$$

b)



c)



$$4x^2 + 9y^2 = 36$$

$$8x + 18y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{8x}{18y}$$

$$\text{at } P_{\text{TAN}} =$$

$$= \frac{-24 \cos \alpha}{36 \sin \alpha}$$

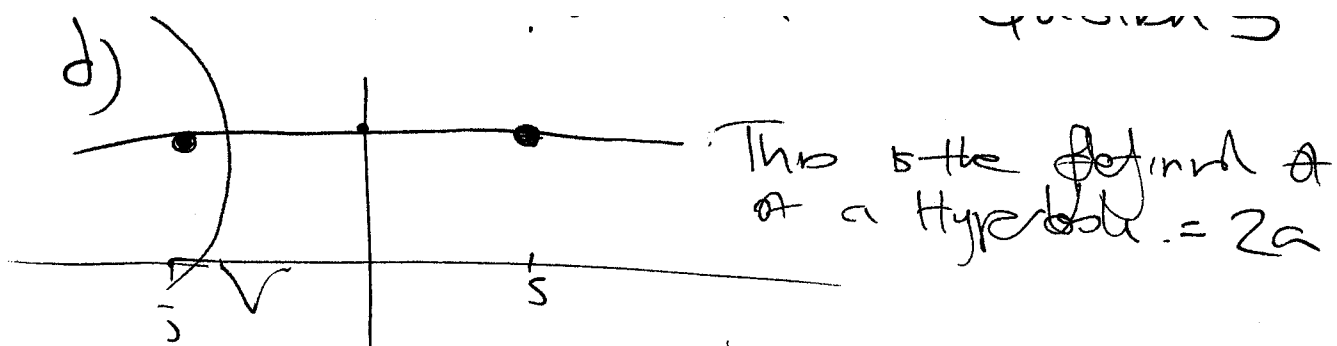
$$\therefore P_{\text{NORMAL}} = + \frac{3 \sin \alpha}{2 \cos \alpha}$$

$$\text{Similarly } Q_{\text{NORMAL}} = \frac{3 \sin \beta}{2 \cos \beta}$$

$$\text{we know } \frac{3 \sin \alpha}{2 \cos \alpha} \cdot \frac{3 \sin \beta}{2 \cos \beta} = -1$$

$$\therefore \frac{9 \tan \alpha \tan \beta}{4} = -1$$

$$\therefore 4 \cot \alpha \cot \beta = -9$$



11) Foci: $(5, 0)$ $(-5, 0)$

$$2ae = 10$$

$$a = 2$$

$$\therefore e = \frac{5}{2}$$

$$b^2 = 21$$

$$\frac{x^2}{4} - \frac{y^2}{21} = 1$$

$$x = 5$$

$$\frac{25}{4} - \frac{y^2}{21} = 1$$

$$\frac{21}{4} = \frac{y^2}{21}$$

$$\frac{21^2}{4} = y^2$$

$$\pm \frac{21}{2}$$

$$\therefore \text{Latus} = \frac{42}{2} = 21$$

$$b^2 = a^2(e^2 - 1)$$

$$b^2 = 4\left(\frac{25}{4} - 1\right)$$

$$\frac{b^2}{4} = \frac{21}{4}$$

~~Same shape~~
~~no need to~~
 show shift.